Homework 2

Due Tuesday, November 3rd at 11:59 PM PT

Homework Instructions

- This homework is worth **40 points** and covers **Lectures 6–10 and Labs 3–5**.
- Write your solutions in **R Markdown**, knit to PDF, and submit the PDF on Gradescope by the deadline above.
- Give complete and concise answers. We **grade both completeness and conciseness**. Explain your results and reasoning without unnecessary detail.
- As a guide, keep written explanations to **70 words or fewer for 2-point parts**, **100 or fewer for 3-point parts**, and **130 or fewer for 4-point parts**. These are maximums, not minimums. Shorter answers are fine if they address every part of the question.
- Include the R code used for every calculation, simulation, and plot, along with relevant outputs. (Code is not needed for purely conceptual parts.)
- **Code, equations, calculations, tables, and plots do not count** toward these word limits.
- On Gradescope, **assign the correct PDF pages to every question and subpart** so your work is easy to find and grade.
- Unless stated otherwise, use $\alpha = 0.05$. You may use base R or tidyverse code.

Introduction

Consider a university reviewing its shuttle service. It has results from a small experiment with an arrival-time display and trip records from a hypothetical shuttle network. The university wants to know what the results tell us and what they do not.

Question 1 uses results from the experiment. **Questions 2–4** use `shuttle_trips.tsv`, a dataset of 280 shuttle trips. Each row has:

- `trip_id`: a trip ID;
- `distance_km`: distance traveled, in kilometers;
- `duration_min`: trip duration, in minutes;
- `stops`: number of stops;
- `period`: Peak or Off-peak;
- `route`: Campus Loop, Medical Center, or North Connector;
- `weather`: Dry or Wet.

In the trip data, the university did not randomly assign the route, weather, or travel period. Be careful about claiming that one variable *causes* another.

Question 1 [10 points]

A university tests whether showing live arrival times reduces how long people wait for a shuttle. The display is randomly turned *On* for 24 shifts and *Off* for 24 other shifts. Each observation is the *average wait during one shift*. Assume the shifts are independent.

Display	Shifts	Mean wait (minutes)	Sample SD
Off	24	10.4	2.0
On	24	8.0	2.1

The university will install the display if it reduces the true mean wait by **at least 2 minutes**.

(a) [2 points] Let Δ be the difference in population mean waiting times (On minus Off). Write H_0 and H_A for testing whether the display changes the mean wait. Estimate Δ and explain what its sign means.

(b) [3 points] A manager says, "The difference is 2.4 minutes, so we already know it is statistically significant." Use the table to check this claim. Find the standard error, a suitable two-sided test statistic, and its p-value. State your conclusion. You may use Welch's t-test or a two-sample t approximation with 23 degrees of freedom.

(c) [3 points] Find a 95% confidence interval for Δ. The manager now says, "The estimated reduction is greater than 2 minutes, so we should install the display." Does the 95% confidence interval provide enough evidence that the true mean reduction is at least 2 minutes? Explain how this differs from finding evidence of any change.

(d) [2 points] A student tries a permutation test. They shuffle the waiting times *within* the On group and *within* the Off group, and calculate the difference in group means each time. What will happen to the calculated differences? Explain why this does not give a useful permutation test, and say what the student should shuffle instead.

Question 2 [10 points]

A planner wants to predict shuttle trip duration and understand how Peak-period duration differences relate to other trip characteristics. Use the trip dataset for this question.

(a) [3 points] Start with a regression using `distance_km` to predict `duration_min`. Report the estimated distance slope, its 95% confidence interval, and the p-value for a zero slope. Explain the slope in minutes per kilometer. A planner says, "A trip twice as long in distance should take exactly twice as many minutes." Does the fitted model say this? Explain.

(b) [4 points] Compare the mean duration of Peak and Off-peak trips in two ways: first without other predictors, then after accounting for distance, stops, route, and weather. Give the estimated Peak minus Off-peak difference and a 95% confidence interval in each case. Why might the two estimates differ? Does the second model prove that Peak hours *cause* longer trips?

(c) [3 points] Use Campus Loop as the reference route in the second model. Find the estimated difference in mean duration between Medical Center and North Connector, for trips with the same other predictors. A planner says, "Both routes have significant coefficients, so their difference must be significant too." Is that correct? Explain how you would test the difference between these two routes. You do not need to calculate its p-value.

Question 3 [10 points]

Use the regression with distance, stops, period, route, and weather from Question 2(b).

(a) [2 points] A student notices that the residuals have mean zero and says, "This shows that the regression assumptions are satisfied." Show why least squares with an intercept makes the residuals sum to zero. Then explain why this does not check the assumptions.

(b) [3 points] Make a residual-versus-fitted plot and one other diagnostic plot. Discuss two things you checked and what you see in the plots. If the residual spread changes with fitted values, what could this mean for the usual confidence intervals and p-values from `lm()`?

(c) [3 points] A researcher wants to study a two-sided 5% test for the slope in simple linear regression. Keep the 40 distances x_i fixed at `seq(1, 10, length.out = 40)`. Generate independent values using

$$Y_i = 6 + \beta x_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 3^2).$$

For each of $\beta = 0$ and $\beta = 0.4$, simulate 2,000 datasets. For each dataset, fit a regression and record whether the test rejects a zero slope. Set and report a random seed. Report the two rejection proportions and explain what each one measures. A student claims, "Running more simulations would increase the test's power." Is the student correct?

(d) [2 points] Every trip has a different `trip_id`. Does that mean trip durations are independent? Give one reason two different trips might have related durations, and suggest one way to collect data that would make independence more reasonable. Would a normal-looking histogram of residuals settle this question?

Question 4 [10 points]

A planner uses the regression from Question 2(b) to compare routes. The route names are *categories*, not measurements. Another student suggests changing how route is recorded to make the model simpler. Keep distance, stops, period, and weather in every regression in this question.

(a) [2 points] The student first creates three columns, one for each route, with 1 if a trip uses that route and 0 otherwise. They put all three columns and an intercept in the regression. Why is this a problem? Give one way to fix it.

(b) [3 points] The student then codes Campus Loop as 1, Medical Center as 2, and North Connector as 3. They fit a regression using this number as an ordinary numeric predictor instead of using `route` as a categorical predictor. What does this assume about the differences between routes? Fit this model and compare its residual sum of squares (RSS) with the model from Question 2(b). Which way of treating `route` makes more sense here? Explain.

(c) [3 points] The planner changes the reference weather from Dry to Wet in the model from Question 2(b). Without refitting, find the new intercept and the coefficient for Dry. Explain what the new intercept means. What happens to the fitted trip durations and the coefficients for the other predictors?

(d) [2 points] In the numeric-route model from part (b), the p-value for the route number is greater than 0.05. A student says, "This proves that route makes no difference to trip duration." Is that conclusion justified? Explain what this p-value actually tests and why it may miss differences between the three routes.